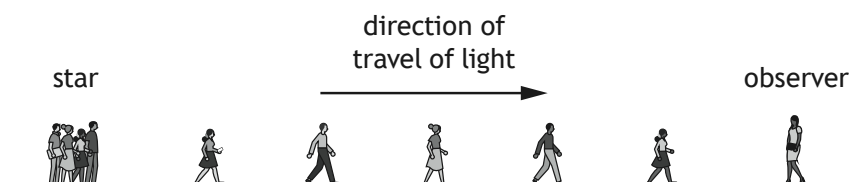


5. A group of students are participating in a demonstration that models the redshift of light from a distant star relative to an observer.

The 'star' is initially stationary relative to the observer.

The students leave the 'star' at intervals of 3.5 s and walk towards the observer.

Each student represents a wavefront of the light emitted by the 'star'.



- (a) The students representing the wavefronts walk with a constant speed of 1.5 m s^{-1} towards the observer.

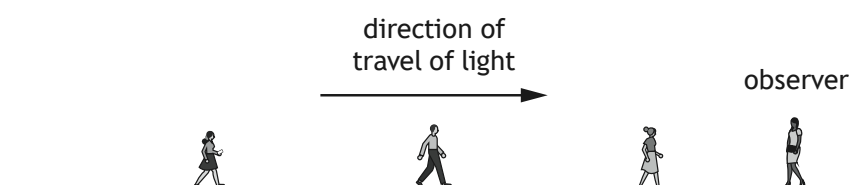
Determine the wavelength of the light represented by the students.

3

Space for working and answer

- (b) The demonstration is repeated, and the 'star' now moves at a constant speed relative to the observer. The wavefronts are still emitted from the 'star' at intervals of 3.5 s.

The pattern now observed is shown below.



State if the 'star' is moving towards or away from the observer.

You must justify your answer.

2



* X 8 5 7 7 6 0 1 1 8 *

5. (continued)

(c) Redshift is evidence for the concept of the expanding Universe. Another piece of evidence for the expansion of the Universe is Olber's paradox.

(i) Explain how Olber's paradox, also referred to as the dark sky paradox, supports the concept of the expanding Universe.

2

(ii) State one other piece of evidence that supports the concept of the expanding Universe.

1

[Turn over



* X 8 5 7 7 6 0 1 1 9 *